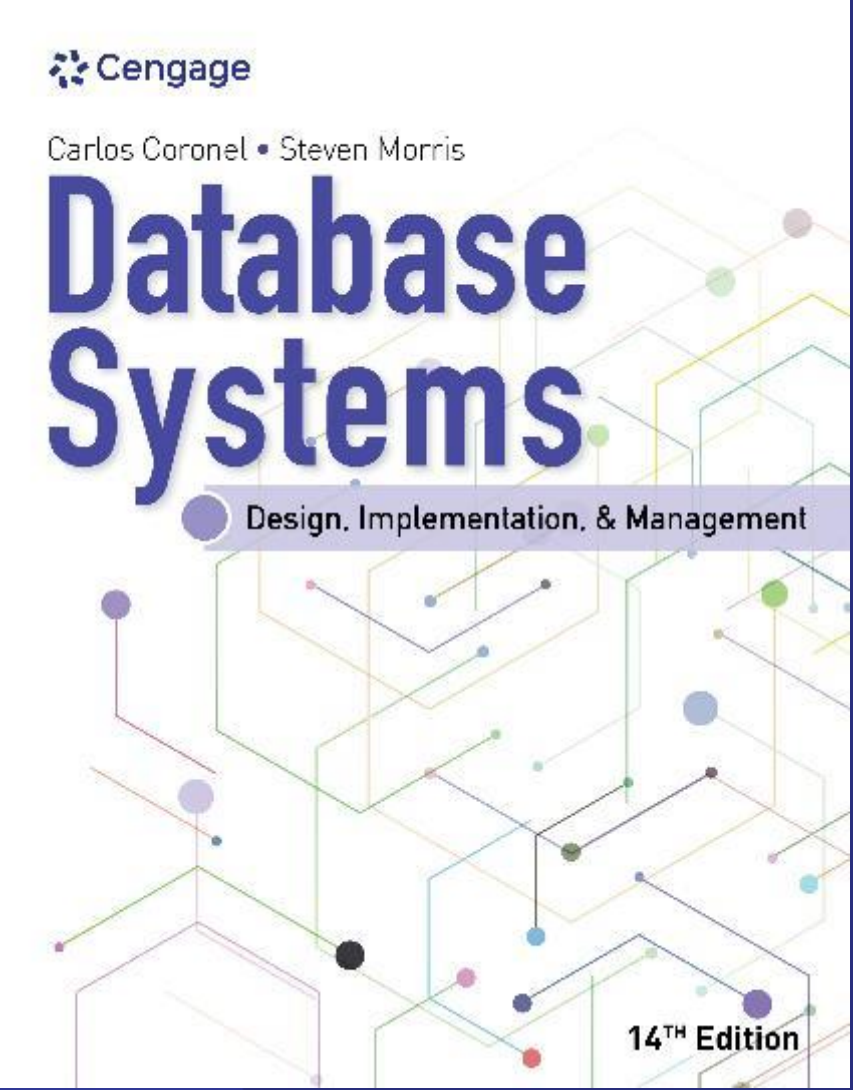
Lecture 9



Database Systems: Design, Implementation, and Management, 14e

Module 5: Advanced Data Modeling

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Chapter Objectives

By the end of this chapter, you should be able to:

1. Describe the main extended entity relationship (EER) model constructs and how they are represented in ERDs and EERDs
2. Use entity clusters to represent multiple entities and relationships in an entity relationship diagram (ERD)
3. Describe the characteristics of good primary keys and how to select them
4. Apply flexible solutions for special data-modeling cases

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The Extended Entity Relationship Model

- The **extended entity relationship model (EERM)** is the result of adding more semantic constructs to the original ER model
  - It is sometimes referred to as the enhanced entity relationship model
- A **diagram** that uses the EERM is called **EER diagram (EERD)**

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Entity Supertypes and Subtypes (1 of 2)

- A university could group employees as faculty, staff, and administrators.
- The **grouping** of employees into **various types provides the following two benefits:**
  - It avoids unnecessary nulls in attributes when some employees have characteristics that are not shared by other employees
  - It enables a particular employee type to participate in relationships that are unique to that employee type
- An **entity supertype** is a generic entity type related to one or more **entity subtypes**
  - The **entity supertype** contains common characteristics
- The entity subtype contains unique characteristics of each entity subtype

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Entity Supertypes and Subtypes (2 of 2)

- Two criteria help the designer determine when to use subtypes and supertypes:
  - There must be different, identifiable kinds of the entity in the user's environment
  - The different kinds or types of instances should each have one or more attributes that are unique to that kind or type of instance

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Specialization Hierarchy (1 of 3)

- Entity supertypes and subtypes are organized in a **specialization hierarchy**, which depicts the arrangement of higher-level entity supertypes and lower-level entity subtypes
  - The relationships are described in terms of “is-a” relationships
- Within a specialization hierarchy, a subtype can only exist within the context of a supertype
  - Every subtype has only one supertype to which it is directly related
- A supertype can have many subtypes

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Specialization Hierarchy (2 of 3)

- A **specialization hierarchy** provides the means to do the following:
  - Support attribute inheritance
  - Define a special supertype attribute known as the subtype discriminator
  - Define disjoint or overlapping constraints and complete or partial constraints

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Specialization Hierarchy (3 of 3)

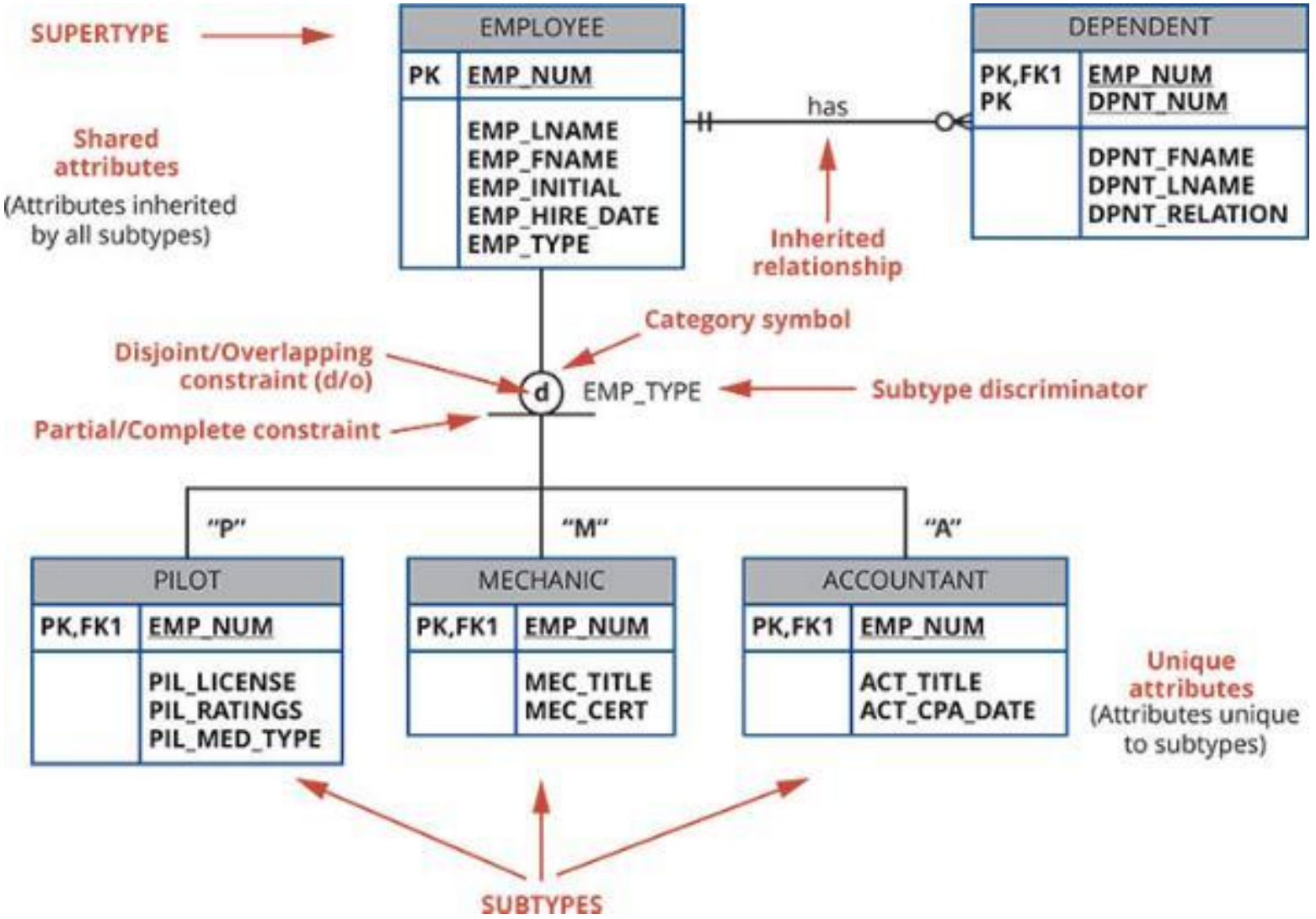


Figure 5.2 A Specialization Hierarchy

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Knowledge Check Activity 5-1

- **What is an entity supertype, and why is it used?**

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Knowledge Check Activity 5-1: Answer

- **What is an entity supertype, and why is it used?**
- **Answer:** An entity supertype is a generic entity type that is related to one or more entity subtypes, where the entity supertype contains the common characteristics and the entity subtypes contain the unique characteristics of each entity subtype. The reason for using supertypes is to minimize the number of nulls and to minimize the likelihood of redundant relationships.

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Inheritance (1 of 2)

- The property of **inheritance** enables an entity subtype to inherit attributes and relationships of the supertype
  - All entity subtypes inherit their primary key attribute from their supertype
- At the implementation level, the supertype and its subtype(s) maintain a 1:1 relationship
- Entity subtypes inherit all relationships in which supertype entity participates
- Lower-level subtypes inherit all attributes and relationships from its upper-level supertypes

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Inheritance (2 of 2)

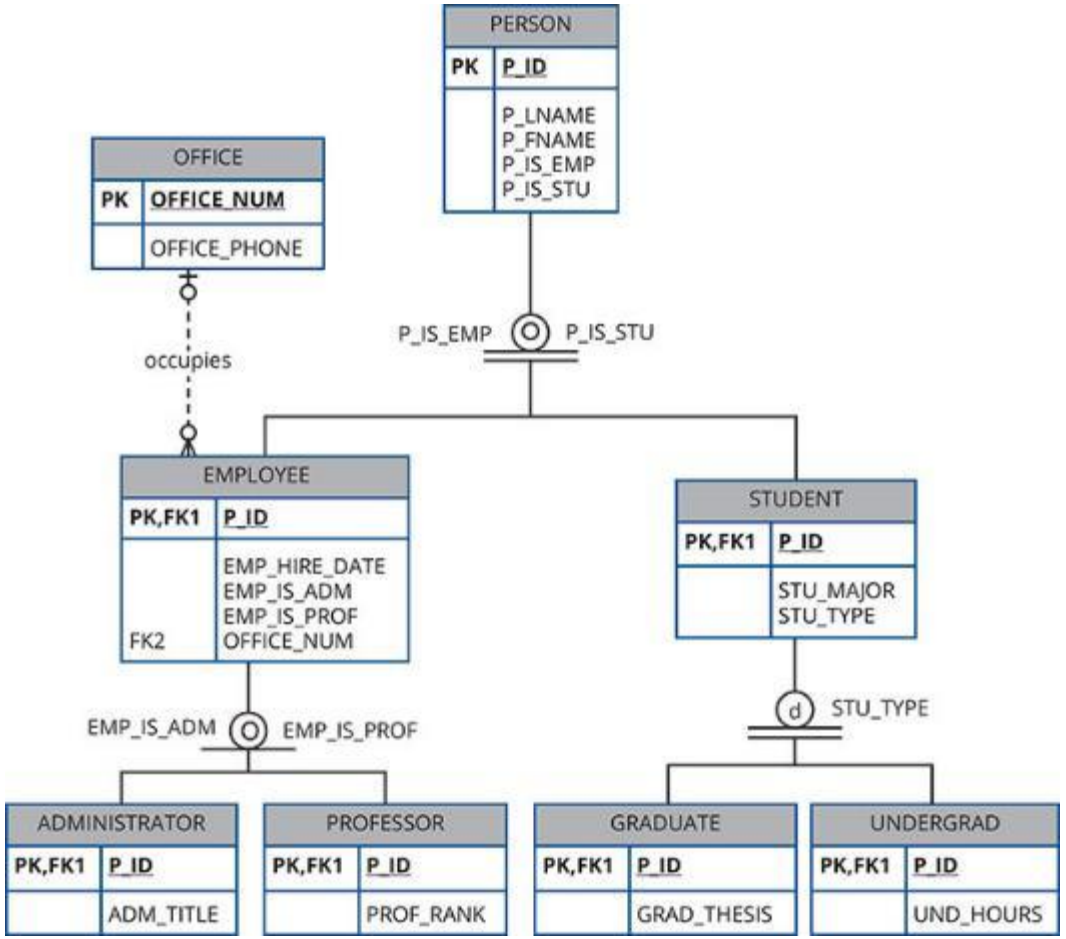


Figure 5.4 Specialization Hierarchy with Overlapping Subtypes

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Subtype Discriminator

- A **subtype discriminator** is the attribute in the supertype entity that determines to which subtype the supertype occurrence is related
- The default comparison condition for the subtype discriminator attribute is the equality comparison
- In some situations the subtype discriminator is not necessarily based on an equality comparison

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Disjoint and Overlapping Constraints (1 of 2)

- **Disjoint subtypes** are subtypes that contain a unique subset of the supertype entity set
  - Also known as **nonoverlapping subtypes**
  - The implementation of disjoint subtypes is based on the value of the subtype discriminator attribute in the supertype
- **Overlapping subtypes** are subtypes that contain nonunique subsets of the supertype entity set
  - Implementation of overlapping subtypes requires the use of one discriminator attribute for each subtype

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Disjoint and Overlapping Constraints (2 of 2)

Discriminator Attributes with Overlapping Subtypes

Table 5.1		
Discriminator Attributes		
Professor	Administrator	Comment
Y	N	The employee is a member of the Professor subtype
N	Y	The Employee is a member of the Administrator subtype
Y	Y	The Employee is both a Professor and an Administrator
N	N	The Employee is not required to be a member of any subtype. Only possible if the hierarchy exhibits the partial completeness constraint (see next section).

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Completeness Constraint (1 of 2)

- The **completeness constraint** specifies whether each supertype occurrence must also be a member of at least one subtype
- **Partial completeness** means that not every supertype occurrence is a member of a subtype
- **Total completeness** means that every supertype occurrence must be a member of at least one subtypes

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Completeness Constraint (2 of 2)

Specialization Hierarchy Constraint Scenarios

Table 5.2		
Type	**Disjoint Constraint**	**Overlapping Constraint**
Partial	Supertype has optional subtypes. Subtype discriminator can be null. Subtype sets are unique.	Supertype has optional subtypes. Subtype discriminators can be null. Subtype sets are not unique.
Total	Every supertype occurrence is a member of only one subtype. Subtype discriminator cannot be null. Subtype sets are unique.	Every supertype occurrence is a member of at least one subtype. Subtype discriminators cannot be null. Subtype sets are not unique.

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Specialization and Generalization

- **Specialization** is the top-down process of identifying lower-level, more specific entity subtypes from a higher-level entity supertype
  - Specialization is based on grouping unique characteristics and relationships of the subtypes
- **Generalization** is the bottom-up process of identifying a higher-level, more generic entity supertype from lower-level entity subtypes
  - Generalization is based on grouping common characteristics and relationships of the subtypes

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Specialization Hierarchy (3 of 3)

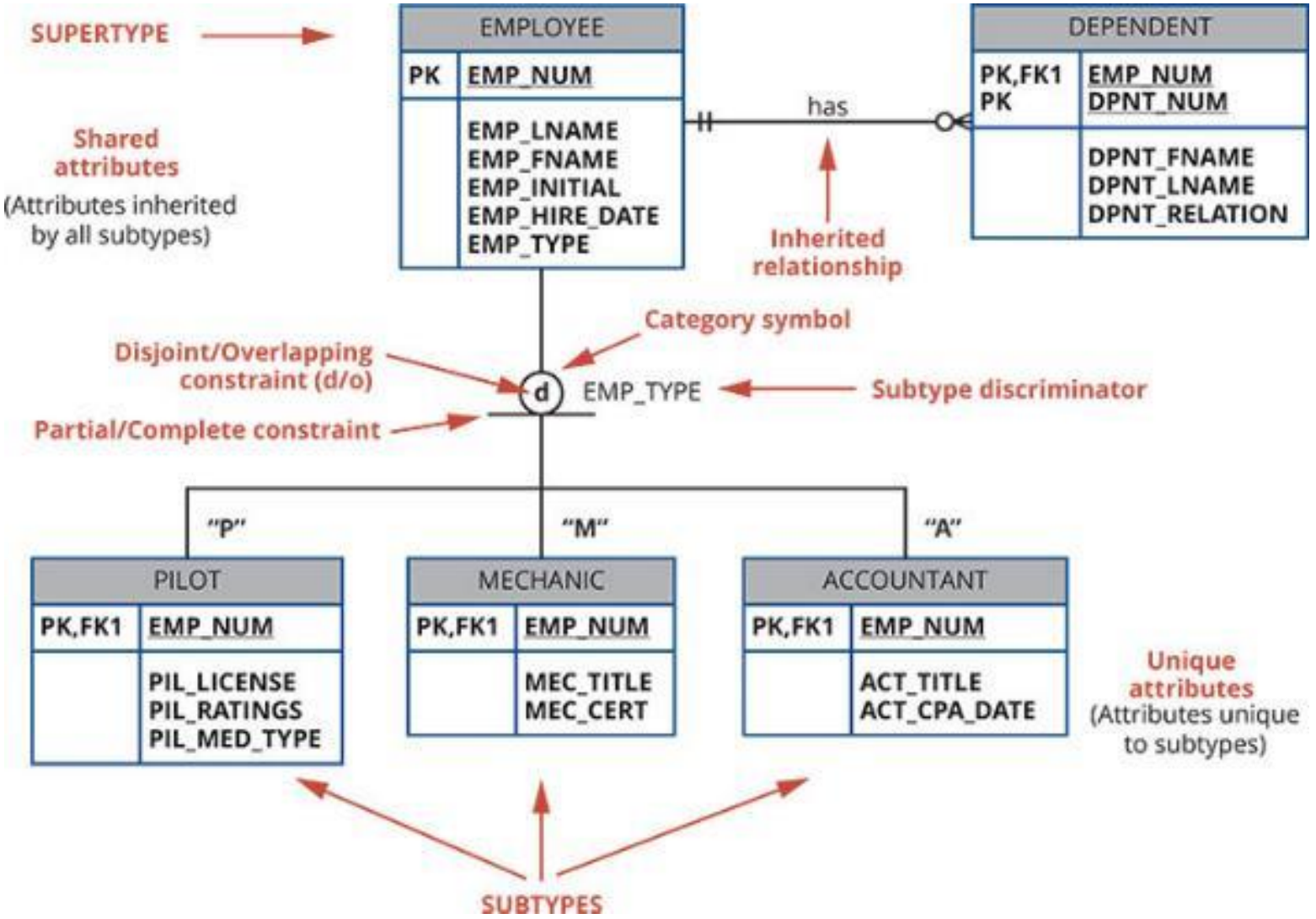


Figure 5.2 A Specialization Hierarchy

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Summary

Now that the lesson has ended, you should be able to:

1. Describe the main extended entity relationship (EER) model constructs and how they are represented in ERDs and EERDs
2. Use entity clusters to represent multiple entities and relationships in an entity relationship diagram (ERD)
3. Describe the characteristics of good primary keys and how to select them
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Concepts

- Supertypes and subtypes
- Specialization Hierarchy
- Inheritance
- Subtype Discrimination
- Disjoint(Nonoverlapping) and Overlapping subtypes
- Completeness/ Partial/ Total Constraints
- Specialization and Generalization

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Questions?



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